EVEN END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech CSE & Biotech — Semester: IV

Name of the Course: Statistical Data Analysis with R — Course Code: TCS-471

Time: 3 Hours — Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	CO1
a)	Why Statistical analysis is required in digital era for giving important insights in strategic decision making? Discuss in brief about uni-variate and bivariate sampling.	
b)	A bag contains 2 red, 3 green, and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is green?	
c)	Discuss the concept of resampling. Also elaborate the various resampling techniques.	
Q2	(20 marks)	CO3
a)	Explain the Bayes Theorem with a suitable example.	
b)	A manufacturer produces light-bulbs that are packed into boxes of 100. If quality control studies indicate that 0.5% of the light-bulbs produced are defective, what percentage of the boxes will contain: (a) no defective? (b) 2 or more defectives?	
c)	Compare and contrast Binomial and Normal distribution taking a suitable example.	
Q3	(20 marks)	CO3
a)	Diabetes = c("Type 1", "Type 2", "Hyper"), Sugar_level = c(90, 150, 350), Age = c(30, 40, 45) You are required to create the data frame of the diabetes data given above, how to find the summary of the data, extract specific column from a data frame and to remove rows and columns in a Data Frame.	

b)	`days <- list("Sun", "Mon", "Tue")`
	`for (x in days)`
	`{`
	`if (x == "Mon")`
	`{`
	`break`
	`}`
	`print(x)`
	`}`
	What will be the output of this R Program code given above ? Modify this code by replacing break with next, what will be the change in the output.	

c)	Create the matrix in R of elements {2, 4, 6, 8, 7, 9, 5, 4, 2} of 3X3. Show the syntax in R for adding additional rows {4,5,7} and column elements {3,4,5} for this existing matrix.	
Q4	(20 marks)	CO2
a)	Discuss the various visualization libraries used in R.	

b) How do you find the levels in R for the given data ?

music_genre <-c("Jazz", "Rock", "Classic", "Classic", "Pop","Jazz", "Rock", "Jazz"))

Which data types in R is used to store and process categorical data?

c) How functions are declared and used in R ? Find all the outputs of the given below program:

```
my_function <- function(country ="Norway") {
paste("I am from", country)
}
my_function("Sweden")
my_function("India")
my_function()
my_function("USA")
```

Q5 (20 marks)

a) Students of same course are studying in two different sections G and H. Six students from section G and eight students from section H have given a test. The marks obtained by them are given below, Is there a significant difference in mean of both the samples using U test. (Take $Z_{.05}=1.96$)

Section G	Section H
50	67
75	63
88	89
64	88
63	74
35	63
	78
	78

b) Describe the various charts and plots available in R along with it's syntax. CO4

a) Box plot

b) Scatter plots

c) Bar charts

d) 3D Pie charts

c) Consider a predefined dataset **mtcars** in R in which you have to establish the relationship using multiple regression among "mpg" as a response variable with "disp", "noofcylinders" and "wt" as predictor variables. Write the procedure in R to find it's relationship using multiple regression and state the difference between Correlation and Regression.